Today's agenda

↳ Quizzes and learnings

AlgoPrep

int x = 20;

→ Semicolon missing

Quiz 1:

Syntax to declare the variable?

a) type variable_name = value; ✓

b) variable_name type = value;

c) variable_name = value;

d) None of the above;

Quiz 2:

↳ Declare an int variable with name y and value 20?

↳ int y = 20;

Quiz 3:

int a = 10;

int b = 20;

System.out.println(~~a+b~~); → 30

10+20

$\boxed{10}$ a $\boxed{20}$ b

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Quiz 4:

→ Correct x & y

```
int a = 20;
int b = 30;
System.out.println(a + " " + b);
```

"20 " + 30 → 20 30

Quiz 5:

← Case Sensitive

```
int temp = 10;
System.out.println(Temp);
```

↳ error

$\boxed{10}$
temp

Quiz 6:

```
→ System.out.println(a);
  int a = 10;
```

→ error

Quiz 7:

```
int a = 20;
→ int a = 30;
System.out.println(a);
```

→ error

$\boxed{20}$
a

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Quiz 8:

```
int x = 20;
System.out.print(x);
x = 40;
System.out.println(x);
```

20 40

40
~~20~~
x

Quiz 9:

```
int a = 10;
int b = 20;
a = b + 30;
System.out.println(a);
```

assignment operator

20

50

~~50~~
~~20~~
a

20
b

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* variable naming rules:

1. Name can contain lowercase, uppercase or digits [0-9] or $ (dollar) or '_' (underscore).

2. First letter of the variable name can't be number.

3. Can't use reserve words as variable name.
   Reserved words: words which are predefined in java. eg: Public, Static, void, int etc.

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Quiz 10:

```
int x = 10;   ✓
int 1y = 20;  ✗✗
int y@b = 30; ✗✗ @ not allowed
```

⇒ ans = 1

↳ How many of above are correct variable names.

Quiz 11:

```
int _y = 10;  ✓
int xmy = 20; ✓
int x a = 30; ✗✗
int y$z = 45; ✓
```

ans = 3

↳ How many of above are correct variable names.

Quiz 12:

```
int static = 60;
System.out.println(static);
```

↳ error

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* Different categories of data

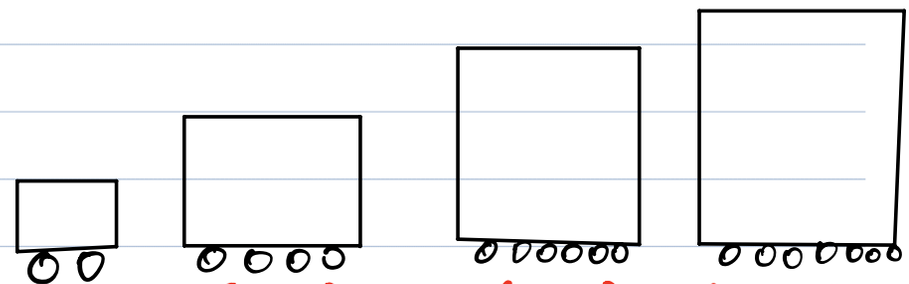
a) Number → Integers, decimal

- ↳ byte
- ↳ short
- ↳ int
- ↳ long

b) Text

c) boolean
true, false

Bhanu Pratap truck service



	(10 kg)	(25 kg)	(50 kg)	(100 kg)
item				
5kg →	✓ (ideal)	✓	✓	✓
75kg →	✗✗	✗✗	✗✗	✓

Range / capacity

byte	-2^7 to 2^7-1	: {-128, 127}	8
short	-2^{15} to $2^{15}-1$	: {-32768, 32767}	16
int	-2^{31} to $2^{31}-1$	: {....., ...}	32
long	-2^{63} to $2^{63}-1$	: {....., ...}	64

↳ Exact explanation of how we got this range is in BITS Class.

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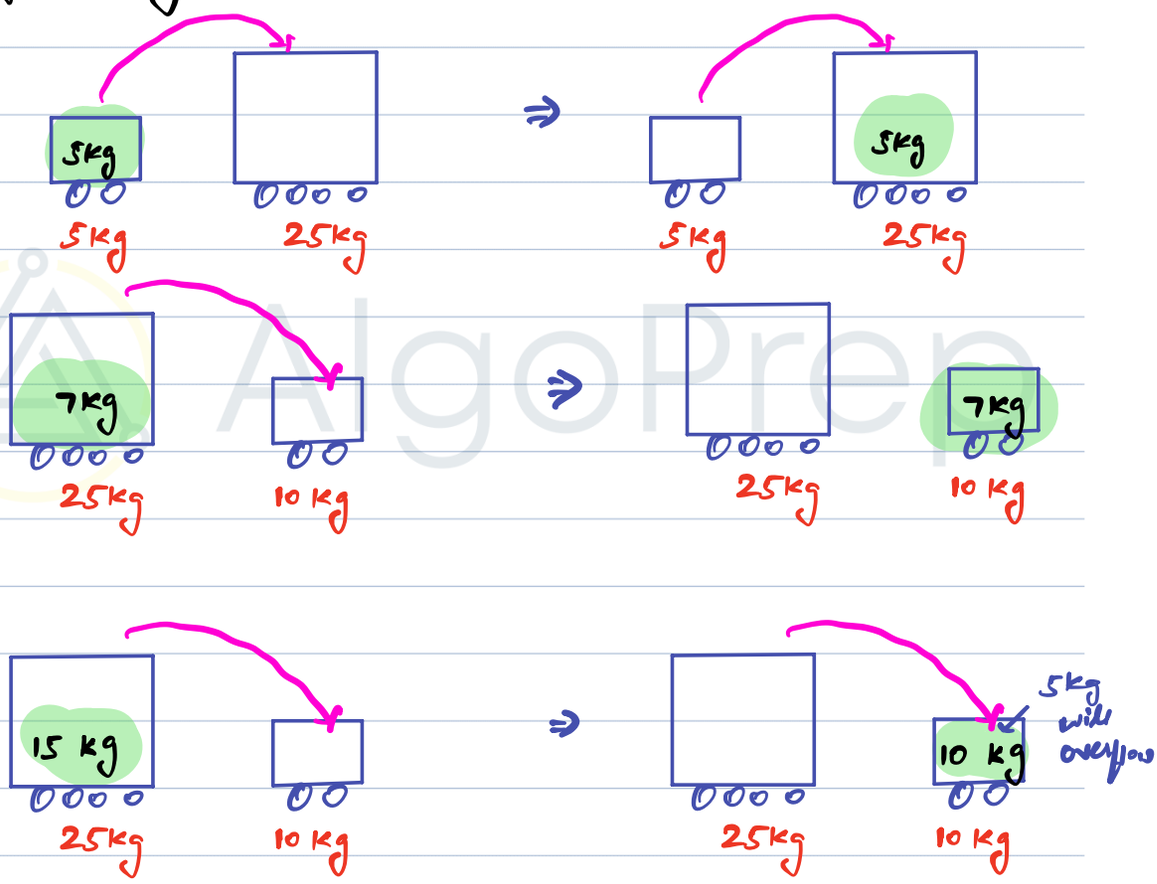
↳ The default value is int in Java.

Break till 9:28 pm

→ Typecasting



implicit (automatic)

explicit (forced)

ex1:

```
int x = 100;
long y = x;
```

→ implicit

ex2:

```
long x = 1000;
int y = (int)x;
```

→ explicit

approx

int : -2^{31} to $2^{31}-1$ → $(-2*10^9$ to $2*10^9)$

long : -2^{62} to $2^{63}-1$ → $(-2*10^{18}$ to $2*10^{18})$

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Quiz 13:

```
int a = 10000;
long b = a;
System.out.println(b);
```

→ implicit

$\boxed{10^4}$ b → 10^4

Quiz 14:

```
long x = 100000;
int y = x;
System.out.println(y);
```

→ explicit

→ int y = (int) x;

explicit typecasting

↳ error

Quiz 15:

```
long a = 10^10L;
int b = (int) a;
System.out.println(b);
```

long → int

10^9

↳ random value

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Quiz 16:

input : 30

```
Scanner scn = new Scanner (System.in);  → error
int n = scn.nextInt();
System.out.println(n);
```

Quiz 17:

input : 30

```
Scanner scn = new Scanner (System.in);  → error
int n = scn.nextInt();
System.out.println(n);
```

Quiz 17:

input : 30

```
Scanner scn = new Scanner (System.in);
int n = scn.nextInt();
System.out.println(n);
```

↳ 30

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input : 24 30

```
Scanner scn = new Scanner(System.in);
int x = scn.nextInt();
int y = scn.nextInt();
System.out.println(x);
```

↳ 24

$\boxed{24}$ x

$\boxed{30}$ y

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